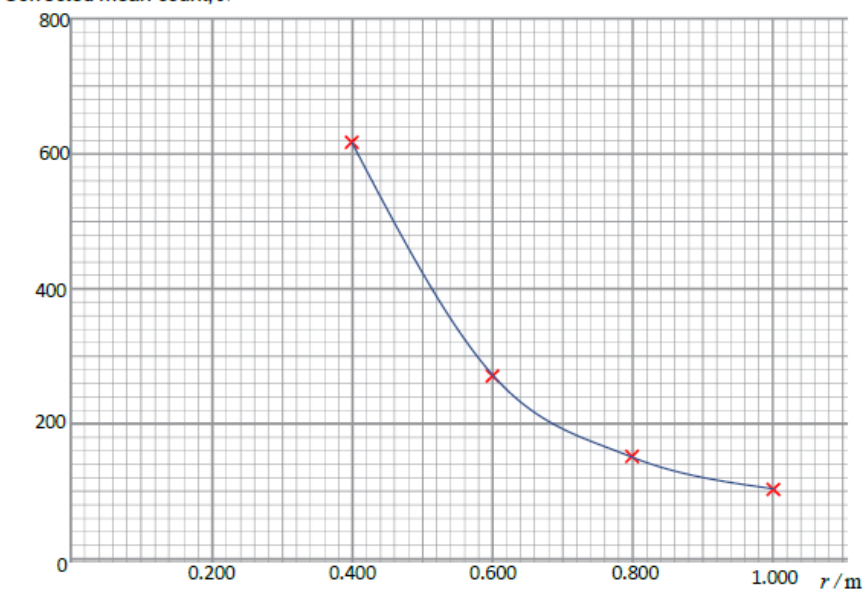


Question			Marking details	Marks available																																																			
				AO1	AO2	AO3	Total	Maths	Prac																																														
5	(a)	(i)	Absorb alpha and beta or only allow gamma through. Accept absorb beta only [due to 40 cm gap]			1	1		1																																														
		(ii)	Background radiation (1) To check for consistency / check for variations (1) Don't accept for a mean			2	2		2																																														
	(b)		<table><tr><th rowspan="2">Distance, r / m</th><th colspan="4">Count</th><th rowspan="2">Uncertainty in mean count</th></tr><tr><th>First reading</th><th>Second reading</th><th>Third reading</th><th>Mean count</th></tr><tr><td>Without source</td><td>19</td><td>22</td><td>25</td><td>22</td><td>3</td></tr><tr><td>1.000</td><td>110</td><td>130</td><td>136</td><td>125</td><td>13</td></tr><tr><td>0.800</td><td>195</td><td>165</td><td>178</td><td>179</td><td>15 or 14 or 16</td></tr><tr><td>0.600</td><td>270</td><td>316</td><td>300</td><td>295</td><td>23 or 21 or 22 or 24 or 25</td></tr><tr><td>0.400</td><td>661</td><td>604</td><td>651</td><td>639</td><td>29</td></tr><tr><td>Without source</td><td>20</td><td>20</td><td>25</td><td>22</td><td>3</td></tr></table> 1 mark per column.	Distance, r / m	Count				Uncertainty in mean count	First reading	Second reading	Third reading	Mean count	Without source	19	22	25	22	3	1.000	110	130	136	125	13	0.800	195	165	178	179	15 or 14 or 16	0.600	270	316	300	295	23 or 21 or 22 or 24 or 25	0.400	661	604	651	639	29	Without source	20	20	25	22	3		2		2	2	2
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	(c)	(i)	<table><tr><th>Distance, r /m</th><th>Corrected mean count, N</th><th>Uncertainty in corrected mean count</th></tr><tr><td>1.000</td><td>103</td><td>16</td></tr><tr><td>0.800</td><td>157</td><td>18</td></tr><tr><td>0.600</td><td>273</td><td>26</td></tr><tr><td>0.400</td><td>617</td><td>32</td></tr></table> 1 mark per column ecf on previous table	Distance, r /m	Corrected mean count, N	Uncertainty in corrected mean count	1.000	103	16	0.800	157	18	0.600	273	26	0.400	617	32		2		2	2	2																															
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				AO1	AO2	AO3	Total	Maths	Prac
		(ii)	Corrected mean count, N  All 4 points correct (2) ecf 3 points correct (1) One smooth curve passing through all points $\pm \frac{1}{2}$ small square (1)		3		3	3	3
		(iii)	Error bars in the corrected mean count are large enough to be shown on the graph [approx. 2 units min], but the uncertainty in r is too small [less than one unit]			1	1		1

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(d)		$\ln 119 = 4.78$ or $\ln 87 = 4.47$ (1) $\ln 103 = 4.63$ or the other extreme (1) $4.78 - 4.63$ or $4.66 - 4.47 = [\text{approx. } 0.15]$ or $\frac{0.31}{2}$ (1) No ecf allowed (since 103 and 16 are given) Award 1 mark for $\frac{16}{103} = 0.15, 0.155, 0.16$ Award 2 marks for statement absolute uncertainty in log = fractional uncertainty in count		3		3	3	3
	(e)	(i)	Taking logs: $\ln(N) = -n \ln(r) + \ln(k)$ (1) (accept $\ln I$ in lieu of $\ln N$) Plotting $\ln(N)$ on the y-axis against $\ln(r)$ on the x-axis or comparison with $y = mx + c$ (1) Gradient = $-n$ (1)			3	3	3	3
		(ii)	Minimum gradient = $\frac{3.94-7.00}{0.50-(-1.30)} = [-]1.69 \pm 0.03$ (1) Maximum gradient = $\frac{3.40-7.00}{0.50-(-1.17)} = [-]2.16 \pm 0.03$ (1) Mean gradient = $[-]1.93 \pm 0.03$ (1) ecf Uncertainty = $\frac{1}{2} (-1.69 - (-2.16)) = 0.20 - 0.28$ (1) ecf accept 0.2 No sig fig penalty		4		4	4	4
			Question 5 total	0	14	7	21	17	21